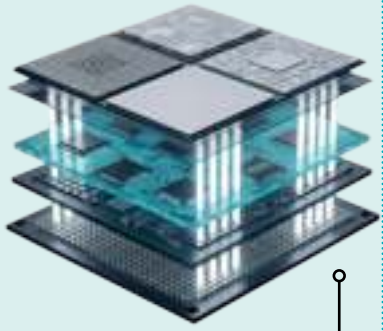
6. PACKAGING



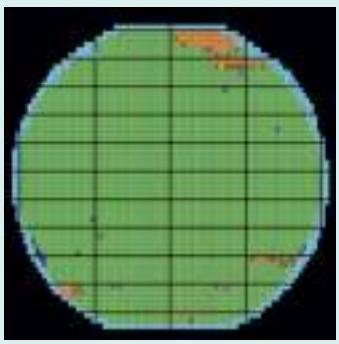
Process: Individual dies are mounted onto substrates, connected via wire bonding or advanced packaging techniques, and sealed

Difficulty: Advanced packaging—including 3D stacking—demands innovative solutions for heat, connectivity and miniaturisation

7. FINAL TEST & BINNING

Process: Packaged chips undergo rigorous electrical, thermal and reliability tests, then are sorted (binned) by performance before shipment

Difficulty: Demands precise stress-testing under varied conditions; performance grading directly affects chip value



times the annual budget for the Jal Jeevan Mission, which provides drinking water via taps to India's 190 million rural households. With the semiconductor market size likely to double from $45-50 billion (Rs 4-4.3 lakh crore) currently to $100-110 billion (Rs 8.87-9.75 lakh crore) by 2030, experts say India may end up spending as much money on importing chips as it does on importing crude oil if it does not start building semiconductor chips indigenously soon.

Nothing brought home the perils of import dependency more forcefully than the years of the COVID-19 pandemic, when the worldwide shortage of chips severely impacted India's auto, electronics and other critical sectors, resulting in costly delays. Given the wildly fluctuating geopolitical landscape amidst US president Donald Trump's policy of 'fall in line or fall behind', the worry is that silicon chip scarcity may turn chronic. And India can scarcely ignore that, since chips are coming to dominate every aspect of our lives, including warfare. This was only too evident during Operation Sindoor, India's response to the Pahalgam terror attack in April, in which autonomous missiles and drones guided by advanced semiconductor processors played a dominant role.

Chips are to the Fourth Industrial Revolution what steel was to its earlier iterations. Even the US, once the undisputed leader in this domain, finds itself being overtaken by East Asian giants such as Taiwan, South Korea and, to some extent, China. To reclaim the American edge, US president Joe Biden in early 2022 launched the Chips and Science Act, sanctioning $52.7 billion (Rs 4.67 lakh crore at current exchange rates) in incentives to domestic manufacturers.

INDIA'S BIG BET

Just a few months earlier, in December 2021, Prime Minister Narendra Modi had unveiled his own high-stakes gambit, when he announced a massive Rs 76,000 crore India Semiconductor Mission (ISM). With the ambitious aim of making India a global hub for semiconductor design and manufacturing, it envisaged building a robust ecosystem and attaining self-reliance in chip manufacturing. As Vaishnaw tells INDIA TODAY in an accompanying interview, "Prime Minister Modi laid down a clear vision that not only should India become self-reliant in chip-making, but we should now become drivers of this bus."

> "The semiconductor mission has made boardrooms across the global chip industry sit up and ask: 'Do I need an India strategy? Am I missing out on a big opportunity in India?'"
>
> **ANKUSH WADHERA** | *MD & Partner, BCG (India)*



It was, by all accounts, an audacious bet. India may be a global software giant, but we were pygmies when it came to making chips that can power industry. Fabricating semiconductor chips is not just another industry, it requires extensive infusions of capital, complex high-precision equipment and tools, advanced technology and skills to etch billions of tiny transistors onto a wafer quarter the size of your fingernail (see graphic *Why Making Semicon Chips is Hard*).

CG Semi's state-of-the-art assembly line, for instance, is an engineering marvel. Cleanliness here borders on obsession, as even a microscopic particle of dust can play havoc with a chip's delicate circuitry. You enter the complex like a surgeon suiting up for an operation